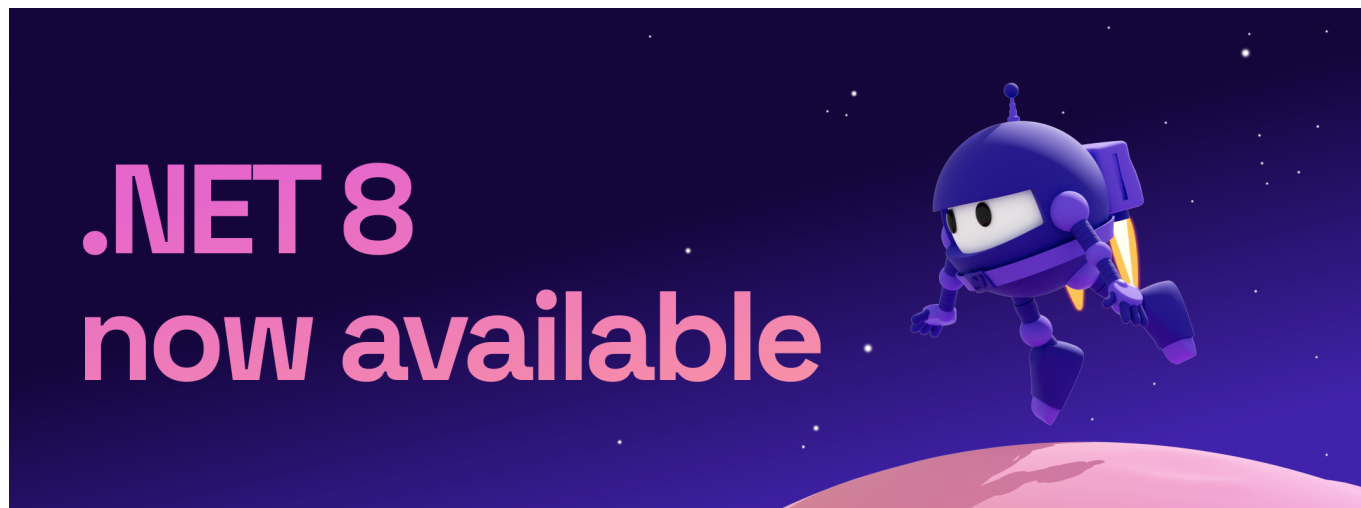


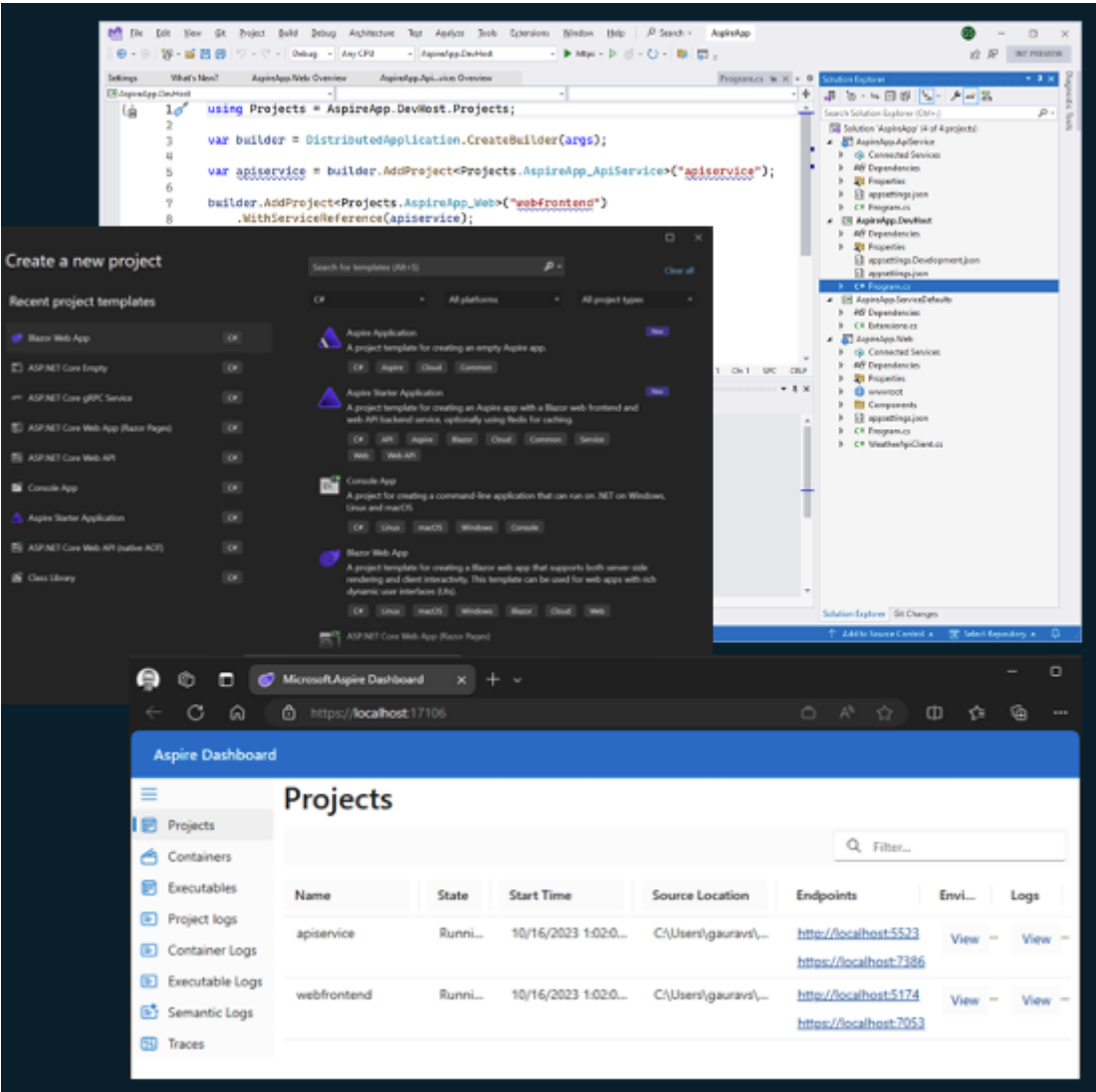


.NET releases are one of the many ways your .NET apps get better over time. Last year we introduced .NET 7, a major release that upgraded your developer experience; enhanced your app's performance; unified the base class library (BCL); provided native compilation and execution of .NET apps on ARM64 systems; and supported the latest version of C# 11.

And starting today, .NET 8 is here and it's packed with updates that help you customize your development and app experience. Let's take a look at what's new.

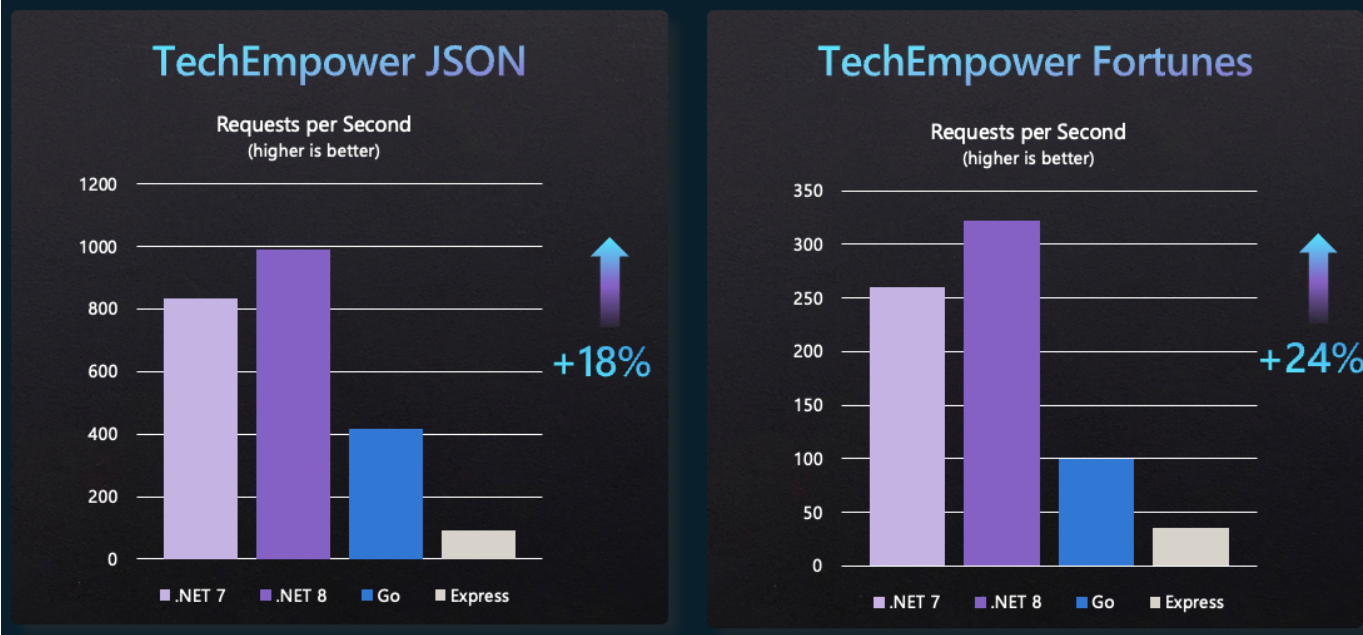
Get Inspired With .NET Aspire (.NET Aspire)

.NET Aspire (Preview) is a new .NET stack for building resilient, observable, and configurable cloud-native applications with .NET. Explore components to handle any cloud-native fundamentals including telemetry, resilience, configuration, and even health checks. Regardless of where you are in your cloud-native journey today, you can easily discover, acquire, and configure new essential dependencies to help you build faster and meet the demands of your growing user base without significant code changes.



Performance Redefined (Performance Enhancements)

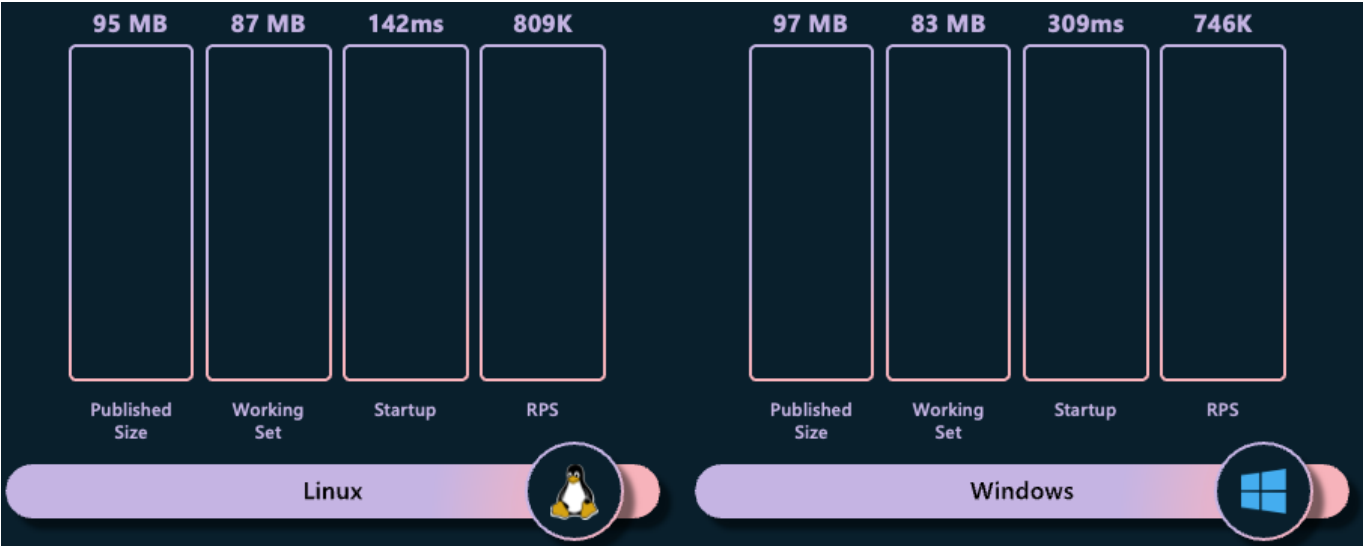
A new code generator called Dynamic Profile-Guided Optimization (PGO) that optimizes your code based on real-world usage is enabled by default and can improve the performance of your apps up to 20%. AVX-512 is now supported which enables you to perform parallel operations on 512-bit vectors of data meaning you can process much more data in less time. The primitive types (numerical and beyond) now implement a new formattable and parsable interface, which enable them to directly format and parse as UTF8 without any transcoding overhead.



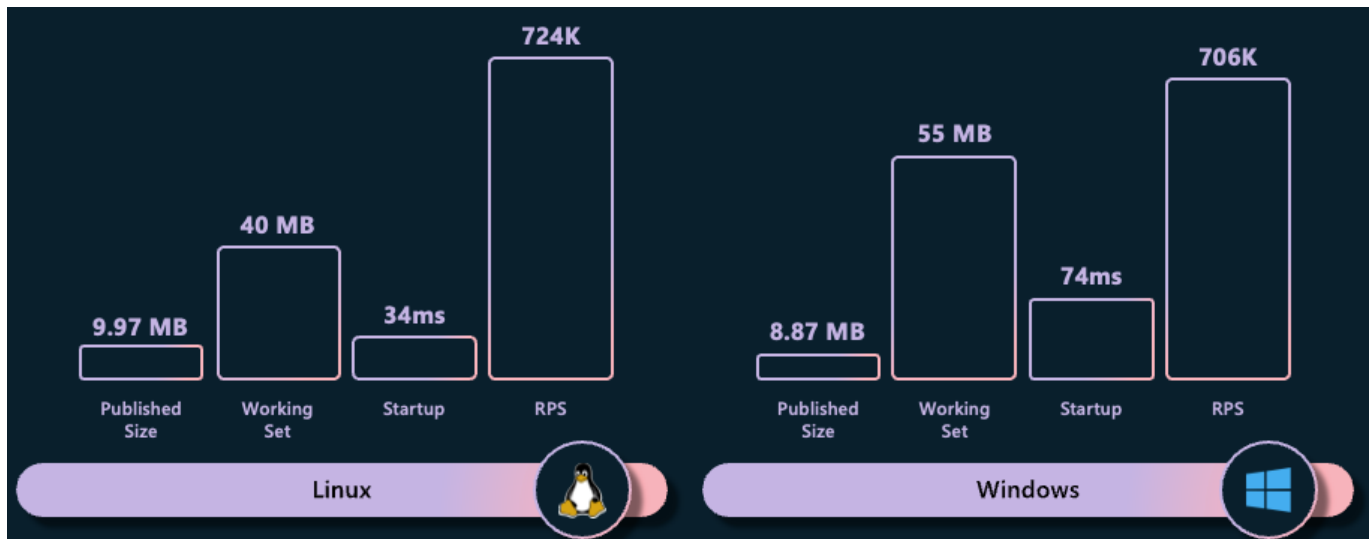
Unlocking Native Speed (Native AOT)

Compile your .NET apps into native code that run faster, uses less memory, and starts instantly. No need to wait for the JIT (just-in-time) compiler to compile the code at runtime; load the JIT compiler and any IL code; provide only the code that is needed for your app; and run in restricted environments where a JIT compiler is not allowed.

Before AOT



After AOT



Coding in Style (C# 12)

C# 12 makes your coding experience more productive and enjoyable. You can now create primary constructors in any class and struct with a simple and elegant syntax. No more boilerplate code to initialize your fields and properties. Be delighted when creating arrays, spans, and other collection types with a concise and expressive syntax. Use new default values for parameters in lambda expressions. No more overloading or null checks to handle optional arguments. You can even use the using alias directive to alias any type, not just named types!

```
// Create an array:
int[] a = [1, 2, 3, 4, 5, 6, 7, 8];

// Create a span
Span<int> b = ['a', 'b', 'c', 'd', 'e', 'f', 'h', 'i'];

// Create a 2 D array:
int[][] twoD = [[1, 2, 3], [4, 5, 6], [7, 8, 9]];

// create a 2 D array from variables:
int[] row0 = [1, 2, 3];
int[] row1 = [4, 5, 6];
int[] row2 = [7, 8, 9];
int[][] twoDFromVariables = [row0, row1, row2];
```

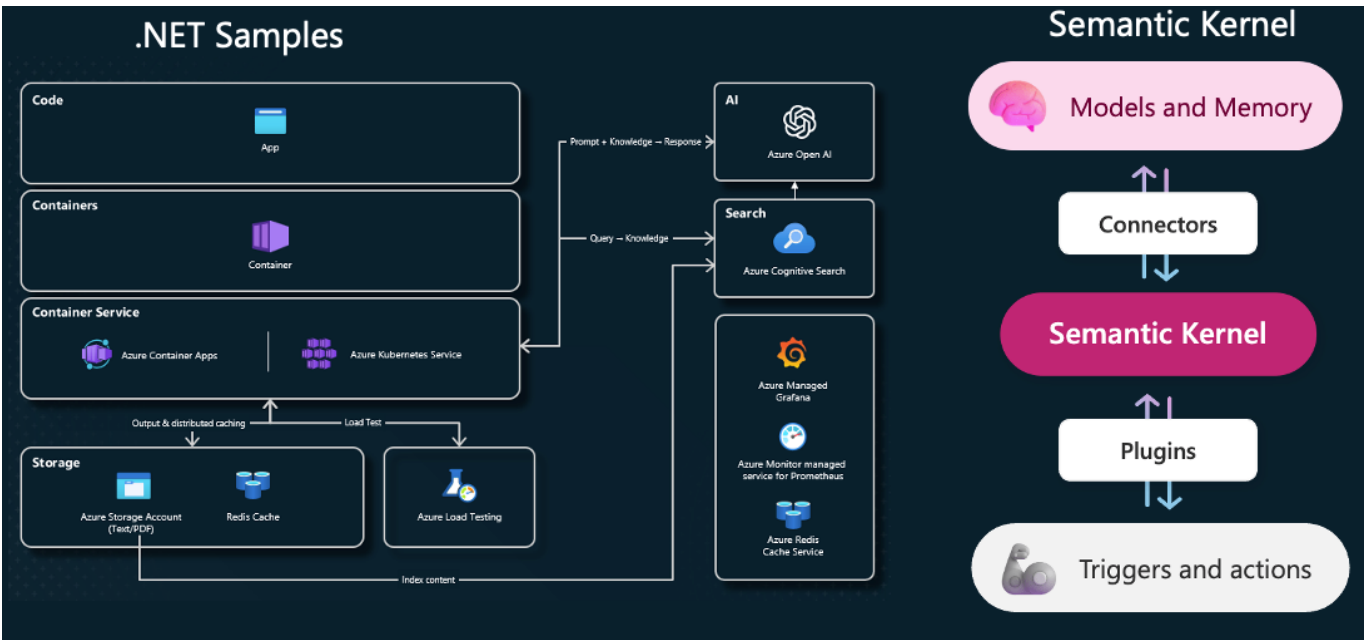
Streamlined Development (Visual Studio Code)

C# Dev Kit is a new collection of extensions that work together to provide a rich and powerful C# environment in Visual Studio Code compatible with Windows, macOS, Linux, and even GitHub Codespaces. The familiar concept of solution management from Visual Studio is now in Visual Studio Code where you can create, open, and manage your C# projects and solutions with a simple and intuitive interface. Build and run your entire solution or individual projects with just one click. Discover, run, and debug your tests with ease using an integrated test explorer. IntelliCode is included to bring you the best of AI-assisted development where you can enjoy whole-line completions, starred suggestions based on your personal codebase, and helps you write cleaner code with style inference and refactoring.



Unlocking Brilliance (AI/ML / Intelligent Apps)

An intelligent app in essence is a piece of software that seamlessly blends cutting-edge technology like AI/ML and human intuition. Leverage new tensor primitives in **System.Numerics** which are optimized for numerical computations, take advantage of hardware acceleration, and abstract concepts like multidimensional arrays simplifying the representation of data and operations in machine learning. When working with your favorite AI-assisted copilot like GitHub Copilot, take advantage of a newly streamlined .NET experience making you more productive, improving your code quality and consistency, and grasp the intricacies between modern .NET and .NET Framework. Semantic Kernel is an open-source SDK that lets you combine AI services like OpenAI, Hugging Face, and more with programming languages like C# helping you create AI apps that combine the best of both worlds.



Blazor Magic (Blazor Unified)

Blazor is a full-stack web UI framework for developing apps. Use a single powerful component model to handle all of your web UI needs, including server-side rendering, client-side rendering, streaming rendering, and progressive enhancement. Partial JIT support is here with the "Jiterpreter" bringing 20% faster UI rendering and 2x faster JSON deserialization by default. The Blazor frameworks gains QuickGrid, sections, routes to named elements, monitoring tools for circuit activity, and improved authentication.

 Full stack web UI	 Enhancements	 WebAssembly
Server-side rendering	Generate static HTML content with components	Multithreading
Enhanced navigation & form handling	QuickGrid	Jiterpreter
Streaming rendering	Sections	Hot reload improvements
Add client interactivity per component or page	Route to named element	SIMD & exception handling enabled by default
Choose component render mode at runtime	Monitor circuit activity	CSP compatibility
	Improved authentication	Webcil packaging

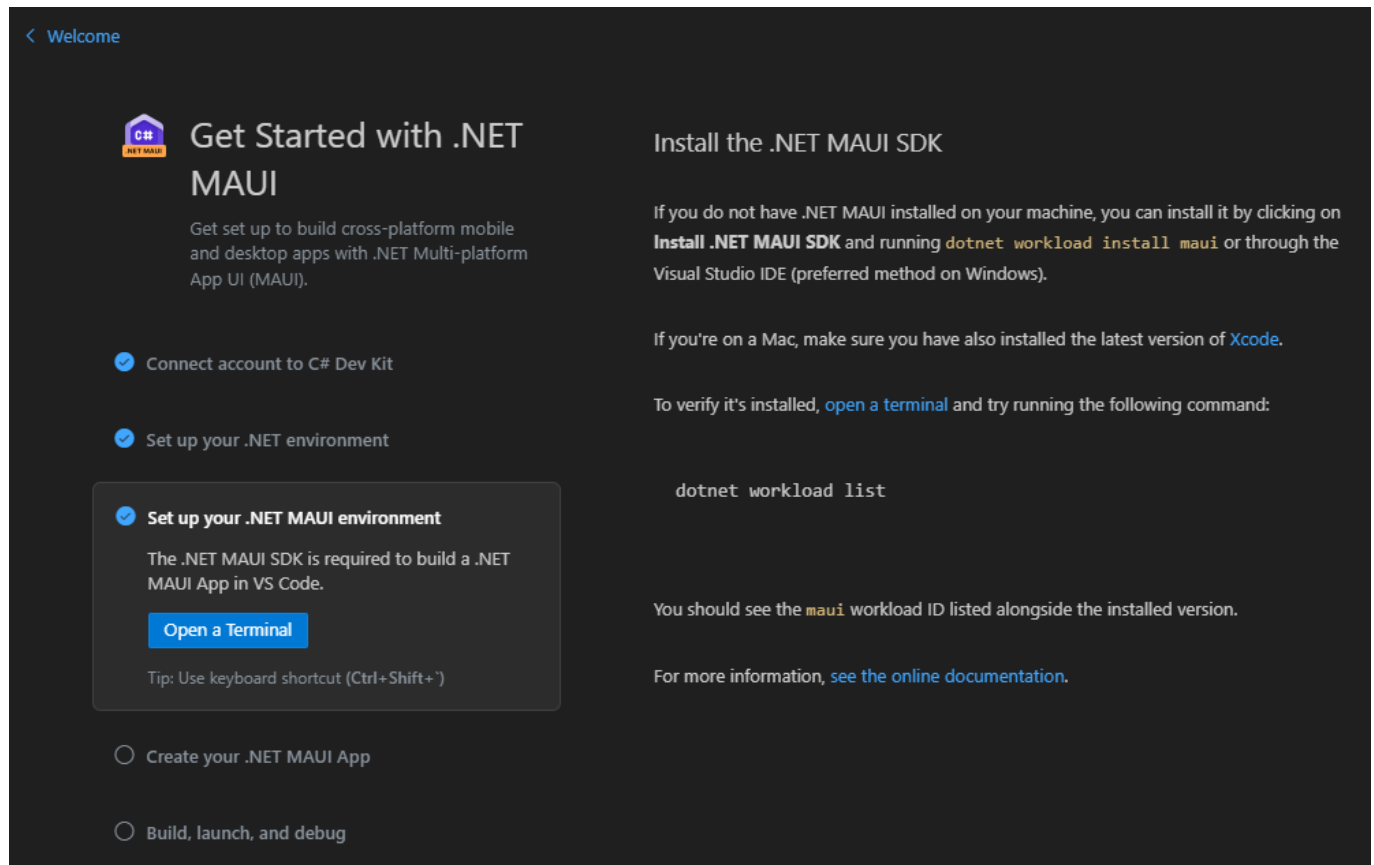
Effortless Containerization (Containers)

Containerizing a .NET application, or any application helps streamline development, deployment, and management processes. A new non-root capability of the Microsoft .NET containers is now the default, which helps your apps stay secure-by-default which include a non-root user. The default container tag is now **latest** to ease your inner development loops. Debian 12 is now used as the default Linux distro in .NET 8 container images. Chiseled Ubuntu images are now available and you can even build multi-platform container images to mix and match the architectures you build.

Hardened	More Productive
Non-Root Base Images	 <code>dotnet publish -r linux-x64 -c Release</code>
USER "app"	"Publish to Container" without Docker/Dockerfile
Default port - 8080	Cross compilation (e.g., build x64 images from arm64)
Mariner distroless	Base non-root image by default
Smaller	Azure Container Registries with Managed Identities
AOT base images	Publishing to local tarball archive
"Composite" base images	Enhanced x-platform support – <code>docker build --platform`& buildx</code>
"extra" base images	
Distroless / Chiseled	

Multi-Platform Excellence (MAUI)

.NET MAUI provides you a single project system and single codebase to build WinUI, Mac Catalyst, iOS, and Android applications. Native AOT now supports targeting iOS-like platforms. A new Visual Studio Code extension for .NET MAUI gives you the tools you need to develop cross-platform .NET mobile and desktop apps. Xcode 15 and Android API 34 are now supported allowing you to target the latest version of iOS and Android. A plethora of quality improvements were made to the areas of performance, controls and UI elements, and platform-specific behavior.



Additional features in .NET 8:

- Streamlined identity in **ASP.NET Core** for single-page applications (SPA) and Blazor providing cookie-based authentication, pre-built APIs, token support, and a new identity UI.
- New "complex types" as value objects, primitive collections, and SQL Server support for hierarchical data is the latest from **Entity Framework Core**.
- Audit your **NuGet packages** in projects and solutions for any known security vulnerabilities.
- Hardware acceleration and support for **OpenFileDialog** comes to **WPF**.
- The **.NET Runtime** adds a new AOT compilation mode for **WebAssembly (WASM)** and **Android**.
- **.NET SDK** revitalizes terminal build output and production-ready defaults.
- **.NET Diagnostics** gets better for Linux and Windows adding enterprise grade scale, performance, and global ready functionality for your telemetry production.
- **Azure** supports .NET 8...
- **Visual Studio 17.8** is packed full of new .NET 8 experiences.
- Go check out **What's new in .NET 8** for everything else!

Get started with .NET 8

For the best development experience with .NET 8, we recommend that you use the latest release of **Visual Studio** and **Visual Studio Code's C# Dev Kit**. Once you're set up, here are some of the things you should do:

- Try the new features and APIs. [Report issues in our issue tracker](#).
- **Test your current app for compatibility.** Learn whether your app is [affected by default behavior changes in .NET 8](#).
- **Test your app with opt-in changes.** .NET 8 has opt-in behavior changes that only affect your app when enabled. It is important to understand and assess these changes early as they may become default in the next release.
- **Update your app with the Upgrade Assistant.** Visual Studio now filters and identifies the specific .NET 8 API changes that are relevant to your app, and walks you through the steps to [upgrade your <TargetFramework> with the Upgrade Assistant](#).
- **Know you're supported.** .NET 8 is officially supported by Microsoft as a [long term support \(LTS\) release that will be supported for 36 months](#).
- **Bonus: eShop Sample for .NET 8.** Follow all the best coding and architecture practices with our [new eShop sample, now updated for .NET 8!](#)

Celebrate .NET 8

//YouTube Placeholder - <https://www.youtube.com/watch?v=VRBF3xvY80w>

- **.NET Conf 2023.** [Join us November 14-16, 2023 to celebrate the .NET 8 release!](#)
- **What's next in .NET?.** [Get involved and learn the latest news on .NET 8](#) and the next version of .NET!
- **Get C# Certified.** [Earn a badge of honor with a freeCodeCamp C# certification](#).

.NET ❤️ Our Community

We would just like to end by saying one big...



Links to work into this blog:

- <https://devblogs.microsoft.com/dotnet/this-arm64-performance-in-dotnet-8/>
- <https://devblogs.microsoft.com/dotnet/debugging-enhancements-in-dotnet-8/>

- <https://devblogs.microsoft.com/dotnet/the-convenience-of-dotnet/>
- <https://devblogs.microsoft.com/dotnet/system-text-json-in-dotnet-8/>
- <https://devblogs.microsoft.com/dotnet/performance-improvements-in-net-8/>
- <https://devblogs.microsoft.com/dotnet/upgrade-assistant-august-2023-update/>
- <https://devblogs.microsoft.com/dotnet/upgrade-assistant-cli/>
- <https://devblogs.microsoft.com/dotnet/upgrade-assistant-general-availability/>
- <https://devblogs.microsoft.com/dotnet/announcing-foundational-csharp-certification/>
- <https://devblogs.microsoft.com/dotnet/debugging-dotnet-containers-with-visual-studio-code-docker-tools/>
- <https://devblogs.microsoft.com/dotnet/new-csharp-12-preview-features/>
- <https://devblogs.microsoft.com/dotnet/check-out-csharp-12-preview/>
- <https://devblogs.microsoft.com/dotnet/improvements-auth-identity-aspnetcore-8/>
- <https://devblogs.microsoft.com/dotnet/why-dotnet/>
- <https://devblogs.microsoft.com/dotnet/securing-containers-with-rootless/>
- PLEASE ADD ANYMORE YOU FIND HELPFUL!